Geophys. J. R. astr. Soc. (1984) 76, 273–287

Palaeomagnetism of granitic intrusives from the Precambrian basement under eastern Kansas: orienting drill cores using secondary magnetization components

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Received 1983 January 21; in original form 1982 September 13

Summary. The Precambrian basement under east-central Kansas was drilled at two circular aeromagnetic positives, one at Osawattamie and one at Big Springs. The core retrieved from these sites is a coarse to medium grained granite which has been dated by U-Pb to be 1350 Ma old. The palaeomagnetism of these azimuthally unoriented cores was studied to see if a technique which uses low-coercivity, low-temperature magnetization components to orient the cores would allow an independent confirmation of the core's mid-Proterozoic age. Orthogonal projection plots of the alternating field (af) and thermal demagnetization data show that the magnetization of these cores is relatively simple, having only two components: a low-temperature, low-coercivity magnetization with steep positive inclinations and a shallow, negative inclination characteristic magnetization for the Osawattamie core or a positive, moderate inclination characteristic magnetization for the Big Springs core. If the declination of the low-temperature, low-coercivity component is aligned parallel to the present field declination, the characteristic directions may be azimuthally oriented. This allows the calculation of palaeomagnetic poles for the Big Springs core (lat. = 4.5°S, long. = 29.9°E) and the Osawattamie core (lat. = 20.2°N, long. = 39.3°E) which are consistent with Irving's apparent polar wander path for Laurentia at about 1300–1400 Ma. Comparison of anhysteretic remanent magnetization (ARM), viscous remanent magnetization (VRM), and isothermal remanent magnetization af demagnetization curves with a natural remanent magnetization (NRM) demagnetization curve suggests that the Osawattamie core probably acquired a piezoremanent magnetization (PRM) parallel to the core axis during drilling.

1 Introduction

A recently completed aeromagnetic survey of Kansas (Yarger 1981) shows a series of very striking circular aeromagnetic highs about 15 km in diameter in north-eastern and central eastern Kansas. In late 1979 and early 1980, the Kansas Geological Survey successfully drilled into the Precambrian basement at two of these aeromagnetic anomalies, one at

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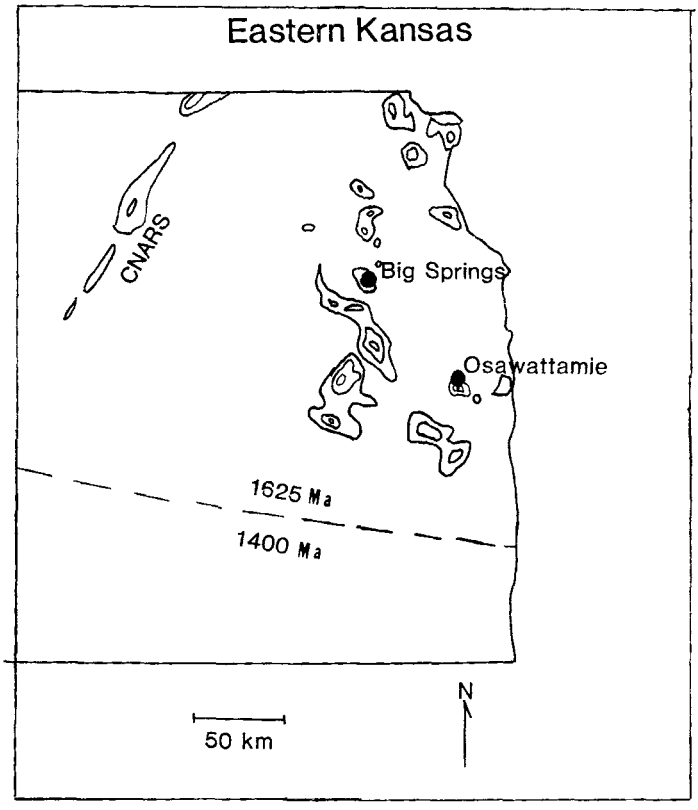


Figure 1. Map, adapted from Yarger (1981), of eastern Kansas showing location of aeromagnetic positives and Kansas Geological Survey drill sites. Dashed line shows postulated boundary between 1625 and 1400 Ma basement rocks. CNARS = Central North American Rift System. All contours show total magnetic field >1600 nT from Yarger's (1981) map. Contour interval 200 nT.

Osawattamie and one at Big Springs, Kansas (Steeples & Bickford 1981) (Fig. 1). The original objectives of this project were to determine if the basement rocks at the aeromagnetic highs were magnetically susceptible enough to produce the observed anomalies, to study the rocks' petrography, and to date the basement radiogenically. Even though unoriented core was collected at these sites, it was hoped that by using the rock's present field magnetic overprints (Van der Voo & Watts 1978) the core could be oriented. If this technique was successful it might allow independent confirmation of the approximately 1350 Ma U-Pb ages of these samples.

2 Geology

The Precambrian basement of Kansas is divided into two terrains; an older terrain of mesozonal granitic rocks (1625 Ma) which underlies the northern half of the state and a younger terrain (1400 Ma) of epizonal granitic and silicic volcanic rocks to the south (Denison *et al.* 1982). The Kansas Geological Survey drill cores suggest that the aeromagnetic highs are caused by granitic, intrusive bodies emplaced in the older, northern terrain. The 1339 ± 12 Ma U-Pb age for the Big Springs core and 1361 ± 6 Ma age for the Osawattamie core (Steeples & Bickford 1981) indicate that these granitic bodies are

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approximately the same age as the southern terrain. Some authors have suggested that the boundary between the southern and northern terrains marks a mid-Proterozoic (~ 1300 Ma) convergence zone (Van Schmus & Bickford 1981). This boundary is marked geophysically by a north-west trending, linear band of aeromagnetic lows (Yarger 1981). The Central North American Rift System (Ocola & Meyer 1973), a possible aborted Keweenawan age rift (~100 Ma), trends south-west–north-east about 120 km to the west of the aeromagnetic highs.

Petrographically, the Osawattamie core is coarse grained with microcline-perthite, plagioclase, quartz, biotite, and some muscovite (Steeples & Bickford 1981). It has 2 per cent magnetite by weight (Steeples 1980). The Big Springs core is medium-grained and mineralogically almost identical to the Osawattamie core, except that it lacks large microcline-perthite grains (Steeples & Bickford 1981).

3 Sampling and measurement

A 7 m long core of Precambrian basement was successfully retrieved at the Osawattamie drill site. The top of this core was drilled at a depth of 658.1 m below the surface. Nine samples (6.4 cm diameter by 5.2 cm long) were provided for palaeomagnetic studies. The nine samples were taken from the core in three groups of three samples; one group from the top of the core, one at a depth of 4.6 m in the core and one at the bottom of the core (Fig. 2). In each three sample group the samples were cut from the core directly adjacent to each other and were given a relative azimuthal orientation. The three separate groups of samples were not oriented azimuthally with respect to each other.

For remanence measurement and demagnetization studies at least two 2.5 cm diameter by 2.5 cm long specimens were drilled from each sample, one for thermal demagnetization and at least one for alternating field (af) demagnetization. If more than two specimens were drilled the additional specimens were af demagnetized.

At Big Springs, Kansas a 3 m long core of Precambrian basement was retrieved from a depth of 905 m below the surface. Seven 8.6 cm diameter by 5 cm thick samples were provided for palaeomagnetic studies. Samples 1, 2, 3 and 4 were taken at relatively evenly spaced intervals from the core and were not oriented azimuthally with respect to each other (Fig. 2). Samples A1, A2 and A3 were cut from a 1.1 m long segment of core between samples 2 and 3 and could therefore be azimuthally oriented with respect to each other. Sample A1 was directly adjacent to sample 2 and A3 was directly adjacent to sample 3. Four specimens were drilled from each sample, two for af demagnetization and two for thermal demagnetization.

In addition to remanence, bulk susceptibility was measured for samples from the Osawattamie and Big Springs cores. One 2.5 cm diameter by 5 cm long specimen from each of the nine Osawattamie samples and from Big Springs samples 1, 2, 3 and 4 were used for these studies.

Rock magnetic studies were conducted on one sample from the Osawattamie core. The sample was given an anhysteretic remanent magnetization (ARM; DC field = 0.05 mT; peak AC field = 100 mT) and was stepwise af demagnetized up to 40 mT. The sample was then demagnetized at 160 mT and allowed to acquire a viscous remanent magnetization (VRM) in a constant field of 0.19 mT for 160 hr. The VRM was progressively af demagnetized in AC fields as high as 10 mT. Finally isothermal remanent magnetization (IRM) acquisition experiments were conducted in which the direct field was increased in steps of 10 mT. The saturation IRM was af demagnetized as well as the IRM acquired at 10 mT.

Remanence measurements were made on both Digico and home-made digitized, fluxgate spinner magnetometers. Thermal demagnetization was conducted with a mu-metal shielded

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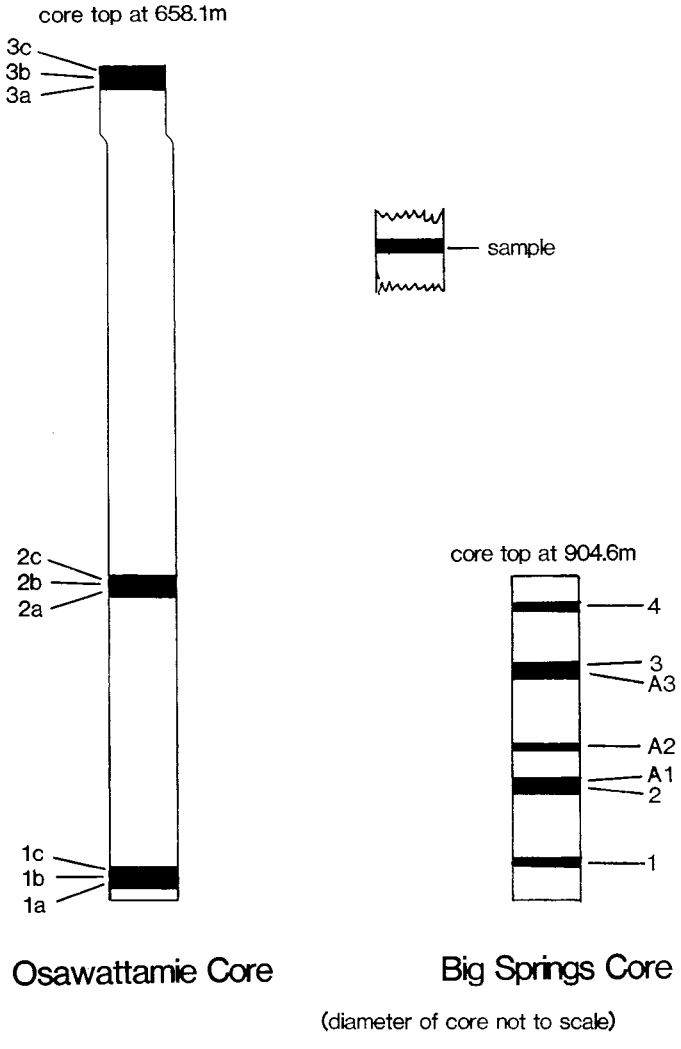


Figure 2. Samples localities in the 7 m long Osawattamie core and in the 3 m long Big Springs core.

oven where the ambient magnetic field was less than 5 nT. Af demagnetization was carried out with both a Schonstedt GSD-1 and a 2-axis, tumbling Inductrol-controlled apparatus. When the tumbling af demagnetizer was used to demagnetize the natural remanent magnetization (NRM) a sample was redemagnetized and remeasured at each demagnetization step after being rotated in the demagnetizer reference frame to average out the effects of rotational remanent magnetization (RRM). In the rock magnetic experiments, a sample was only demagnetized once at each demagnetization step. Susceptibility measurements were made with a Bison susceptibility bridge.

4 Magnetization

All magnetizations were measured in an arbitrary, horizontal reference frame; however, since the cores were drilled within 5° of the vertical (D Steeples, private communication) the

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Table 1. Bulk susceptibility and NRM intensity data for the Osawattamie and Big Springs cores.

Sample	Bulk Susceptibility x 4 π (SI units)	NRM Intensity (A/m)
Osawattamie Core		
1a	3.146-3	5.82
1b	2.400-3	2.60
1c	2.676-3	3.69
2a	2.988-3	3.00
2b	1.907-3	1.75
2c	2.869-3	2.80
3a	2.944-3	2.16
3b	3.576-3	2.29
3c	2.118-3	1.18
Big Springs Core		
1	3.936-3	8.83-1
2	2.209-3	1.81-1
3	1.044-3	2.78-1
4	3.622-3	7.20-1

vertical axis of each sample was known. The orthogonal projections (Zijderveld 1967) that were used to analyse the sample's demagnetization behaviour were plotted in this arbitrary frame. Characteristic directions were calculated by vector subtraction or by a least squares fit of points defining a univectorial decay to the origin. The directions of the secondary magnetic components were calculated by vector subtraction.

The susceptibility and NRM intensity data have relatively high values for these granitic intrusives (Osawattamie, $K = 0.0344$ SI units, NRM = 2.81 A m^{-1}; Big Springs $K = 0.0340$ SI units, NRM = 0.516 A m^{-1} (Table 1). Average Koenigsberger ratios for the Osawattamie and Big Springs material are 2.05 and 0.38 respectively. The unoriented NRM directions for the Big Springs core have inclinations that range from nearly vertically downward to +45° whereas the Osawattamie's NRMs are almost all vertically downward (Fig. 3).

Alternating field demagnetization of the Osawattamie samples does not show the complex multivectorial behaviour typical for rocks of this age. The magnetization of these samples is dominated essentially by two components, a low coercivity component which has steep positive inclinations and the characteristic magnetization which has shallow, negative inclinations (Fig. 4). The steep, low coercivity magnetization is generally removed by demagnetization fields of approximately 30 mT; the characteristic magnetization is removed by fields between 40 and 80 mT and is isolated only after 95–99 per cent of the NRM is removed (Fig. 4). These two magnetizations are seen consistently in almost every sample. Some specimens (five out of 14) show an intermediate coercivity magnetization (20–40 mT) but this magnetization varies in direction from specimen to specimen (Fig. 4). Occasionally samples contain a very low coercivity component of magnetization (Fig. 4).

Thermal demagnetization was only able to remove the steep, positive inclination NRM directions from three of the nine Osawattamie samples. In these three cases (one sample from each group of three samples) the high temperature (> 525°C) characteristic directions agreed with the sample's af demagnetization characteristic directions giving shallow, negative inclinations. These samples lost all of their magnetization at about 580°C. Low temperature

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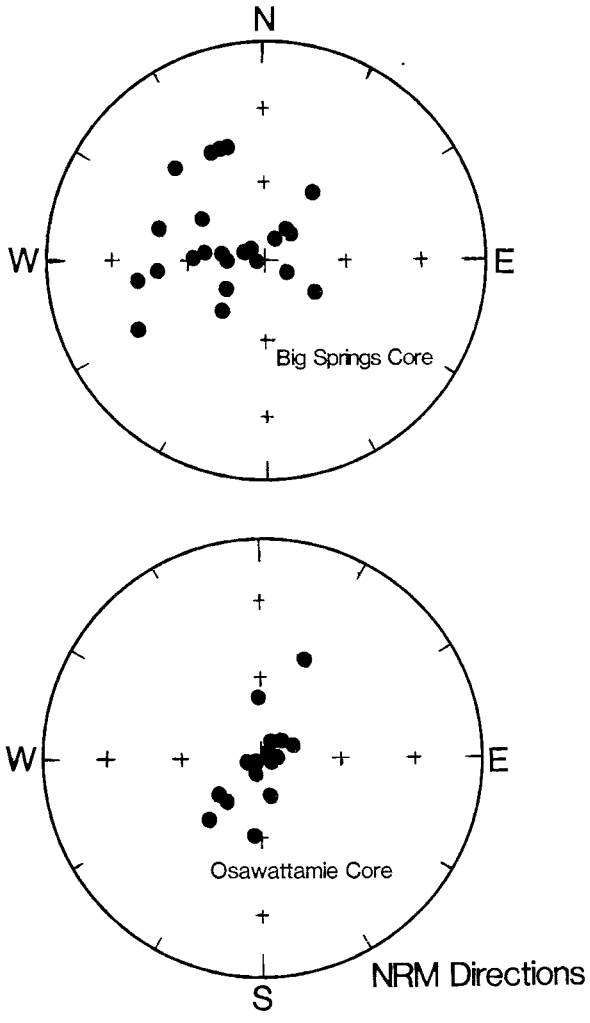


Figure 3. NRM directions measured in arbitrary reference frames. Note that the Osawattamie core has much steeper NRM directions than the Big Springs core.

components removed between 100 and 300°C for eight samples agreed in direction with what had been identified as their low coercivity magnetizations.

In each group of three samples which had relative azimuthal orientations there was directional agreement between the low temperature and low coercivity components within a sample ($K \sim 40.2$; $\alpha_{95} \sim 11.6°$ and between the samples ($K \sim 72.8$; $\alpha_{95} \sim 11.4°$). The thermal and af characteristic directions between the three samples in a group also agreed. The same declination difference between the mean characteristic direction and the mean low-coercivity, low-temperature component in a group was observed for each of the three groups. The high-temperature, high-coercivity characteristic components have declinations which are about 30° clockwise from the low temperature–low coercivity component declinations (Fig. 5).

Af demagnetization of the Big Springs core produced relatively simple orthogonal projections (Zijderfeld 1967) in which low coercivity and high coercivity components of

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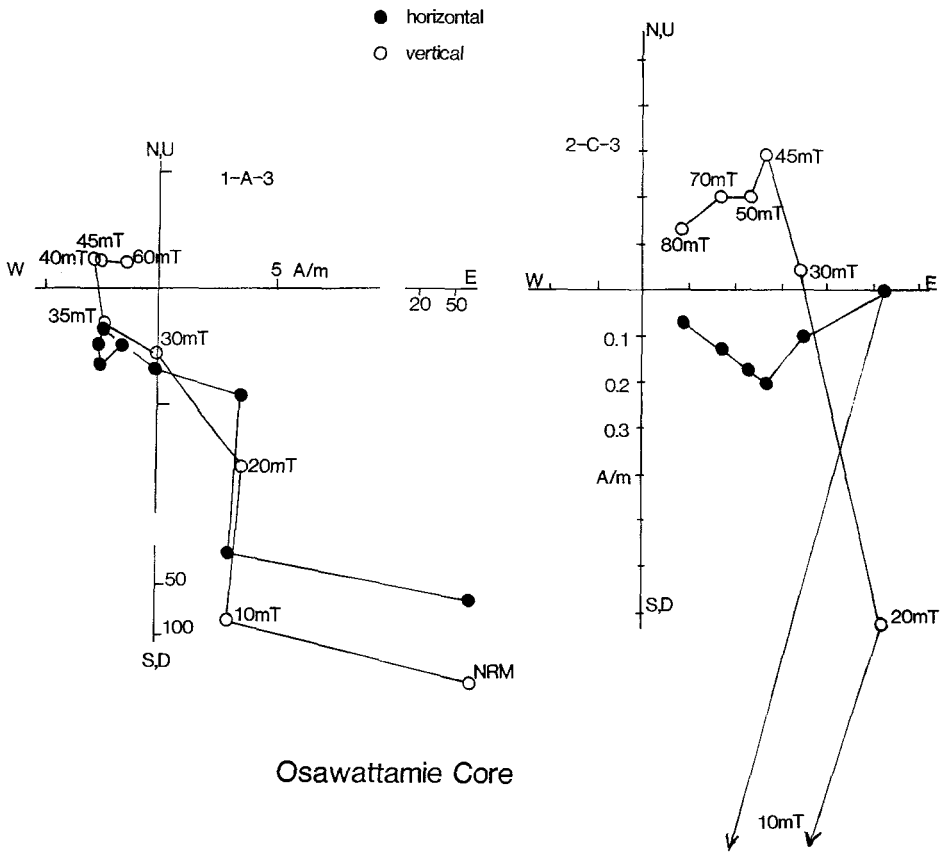


Figure 4. Orthogonal projections (Zijderveld 1967) of af demagnetization data from the Osawattamie core. Sample 1-A-3 shows a very low coercivity component from 0 to 10 mT, probably acquired during storage, the present field component removed between 10 and 20 mT, an intermediate coercivity component, removed between 20 and 40 mT, which does not represent a direction consistently seen in all samples and the characteristic magnetization removed from 45 to 60 mT. Sample 2-C-3 shows only the present field component (20–45 mT), the characteristic magnetization (45–80 mT) and a storage field (<20 mT). Azimuthal orientation of these samples is arbitrary. Note low, negative inclinations for characteristic directions.

magnetization are evident (Fig. 6). The high coercivity component is the characteristic direction and has positive inclinations of about 45–50°. It is removed by af fields from 10 to 60 mT and comprises only 1–5 per cent of the NRM. The low coercivity component generally has steeper positive inclinations (65–80°) and is removed by fields less than 10 mT. In some samples a very low coercivity component (specimen A-3-1 and A-1-1, Fig. 7) was removed by fields smaller than 2.5 mT.

Thermal demagnetization produced results similar to af demagnetization for the Big Springs core. The characteristic directions were removed by temperatures from 300 to 575°C and had +45° to +50° inclinations. The af and thermal characteristic directions agreed within one sample (see specimens 1-C-1 and 1-B, Fig. 7). A steep-inclination, low-temperature component was also evident in these samples. This low temperature component was removed by temperatures <300°C and had the same direction as the low coercivity magnetization in a sample (specimens 1-C-1 and 1-B, Fig. 6). The Big Springs samples lost all their magnetization when heated to temperatures above 580°C (Fig. 7).

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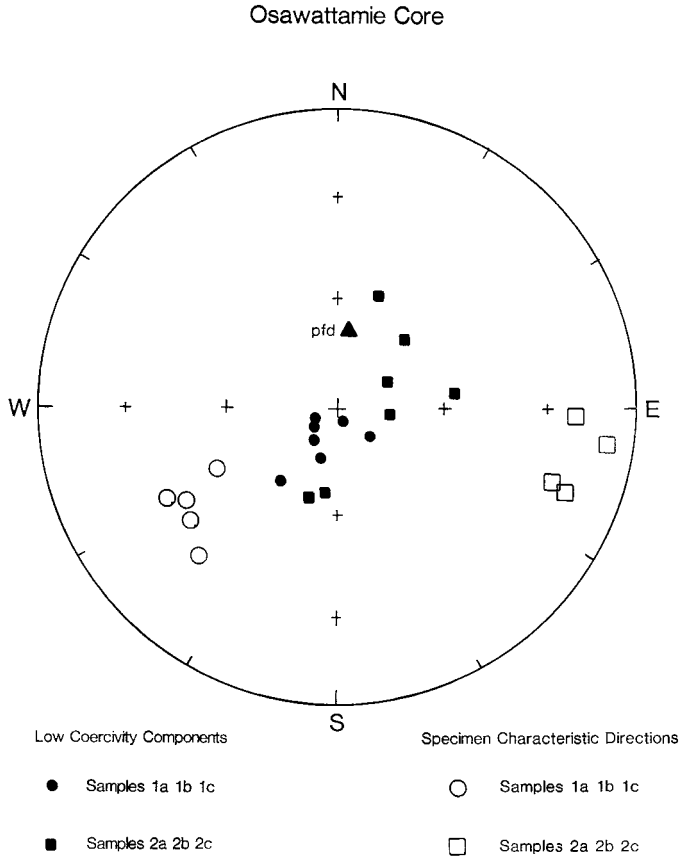


Figure 5. Relationship between low-coercivity magnetization components and characteristic directions for sample groups from the Osawattamie core. Note that the characteristic magnetizations have declinations about 30° clockwise from the low coercivity components. Open symbols are upper hemisphere, closed symbols are lower hemisphere, pfd = present field direction.

The rock magnetic studies of the Osawattamie core show clearly that the af demagnetization characteristics of the NRM are significantly different than those of the ARM and the VRM (Fig. 8). The acquisition of VRM occurred at a slower rate from 0 to 15 hr than at 15–160 hr (Fig. 8). The IRM experiments indicate that the sample saturated in a direct field of 50 mT and that the 10 mT IRM and the saturation IRM were less resistant to af demagnetization than the NRM, but similar in character (Fig. 8).

5 Palaeomagnetic poles for the Osawattamie and Big Springs cores

Since these cores were not azimuthally oriented the only information about the location of the palaeomagnetic poles for these rocks would be that they lie on small circles centred on the sampling sites with radii calculated from the cores' average inclination (e.g. Kean & Swingen 1976). Since simple arithmetical averaging of inclinations tends to underestimate the inclination (Peirce 1976), the mean inclination values used for the small circle calculations (Big Springs core (+49°) and the Osawattamie core (−32°)) were obtained using the method of Briden & Ward (1966). The small circles calculated intersect Irving's (1979)

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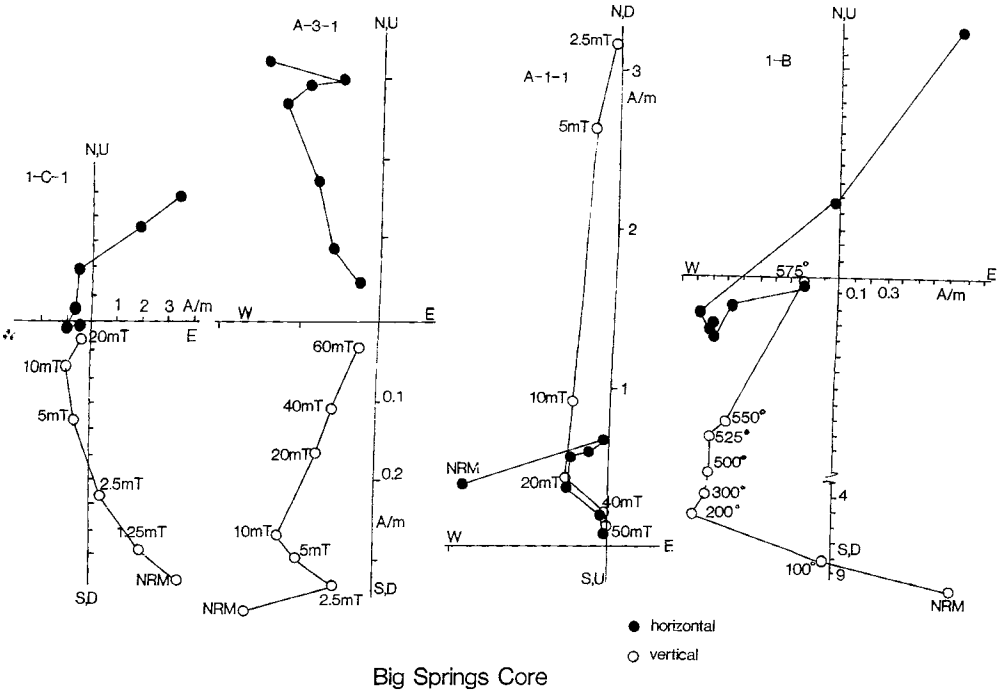


Figure 6. Orthogonal projections (Zijderveld 1967) for demagnetization data from the Big Springs core. Both thermal and af data show the same horizontal component relationship between low-temperature ($20°C < T_B < 200°C$), low-coercivity ($2.5\,mT < H_{CR} < 10\,mT$) magnetizations and characteristic magnetizations, i.e. the present field magnetizations have declinations which point >90° clockwise from the final directions.

apparent polar wander (apw) path for Laurentia at times which correspond to the U-Pb ages for the cores (Fig. 10). Since the apw in the mid-Proterozoic loops back on itself four times there is an obvious ambiguity in this result.

In order to maximize the information contained in the data the low-coercivity, low-temperature components isolated during demagnetization were assumed to be present field

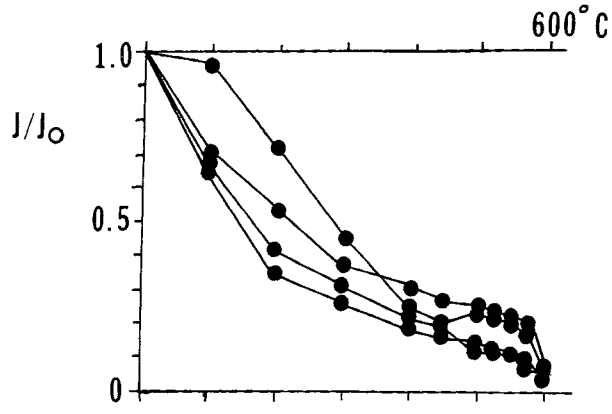


Figure 7. Decrease in magnetization during thermal demagnetization of four representative samples from the Big Springs core. Note that the samples lose all the magnetization by 575°C, suggesting magnetite is the main magnetic carrier.

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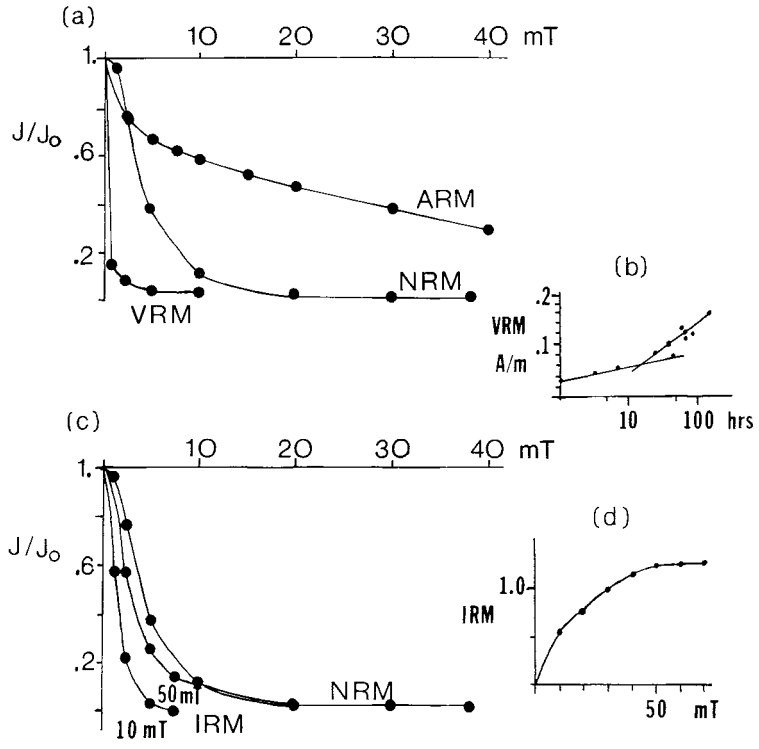


Figure 8. (a) Decrease in intensity of ARM, NRM and VRM with af demagnetization for a sample from the Osawattamie core. Note that the NRM is probably neither a TRM (ARM) nor a VRM. (b) Viscous acquisition for the same sample as in (a) the sample acquires VRM faster after the first 15 hr. (c) Comparison of af demagnetization curves of an NRM and IRM acquired in a 10 mT field and saturation IRM acquired in a 50 mT field for the Osawattamie sample. (d) IRM acquisition curve for the Osawattamie sample.

direction ($I = 68°$, $D = 8°$) magnetic overprints (Van der Voo & Watts 1978). Using this procedure the cores may be oriented in the horizontal plane.

As pointed out in the previous section of the low temperature–low coercivity mean direction for each relatively oriented group of three samples in the Osawattamie core has a declination approximately 30° less than the declination for the group's mean characteristic direction. When the declination of the low-temperature low-coercivity component for each group is aligned parallel to the present field declination it is possible to orient the characteristic directions for this core and arrive at a mean direction of $I = -31.9°$, $D = 44.8°$ ($K = 24.3$, $\alpha_{95} = 9.5°$, $N = 9$, Fig. 9). This mean direction has parameters which agree closely with estimates based on the method of Briden & Ward (1966) ($I = -32°$, $K = 22$) calculated from inclination data. The direction corresponds to a palaeomagnetic pole of lat. = 20.2°N, long. = 39.3°E ($\delta m = 10.7°$, $\delta p = 9.0°$) (Fig. 10).

The use of this technique to orient the Big Springs core was not as straightforward. Samples 1, 2, 3 and 4 from the Big Springs core were demagnetized within 4 months of the core's drilling. Sample 3's low-temperature, low-coercivity magnetization had very low inclinations, so it was assumed that it could not have been acquired parallel to the present field at Kansas. Samples 1, 2 and 4 had high, positive inclination low-temperature, low-coercivity magnetization components. When the declination of these magnetizations were

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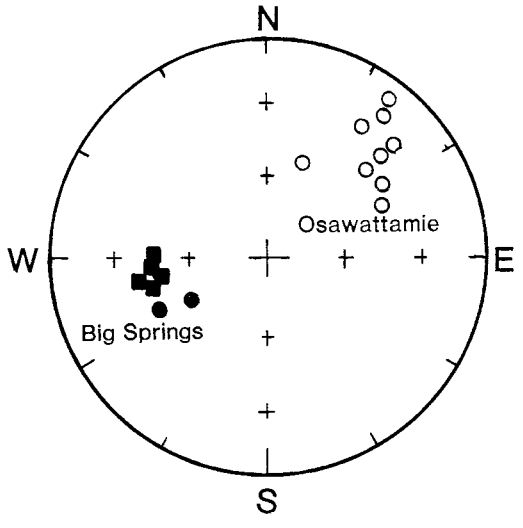


Figure 9. Characteristic magnetizations azimuthally oriented by aligning the declination of low-field, low-temperature magnetizations parallel to the declination of the present field at Kansas.

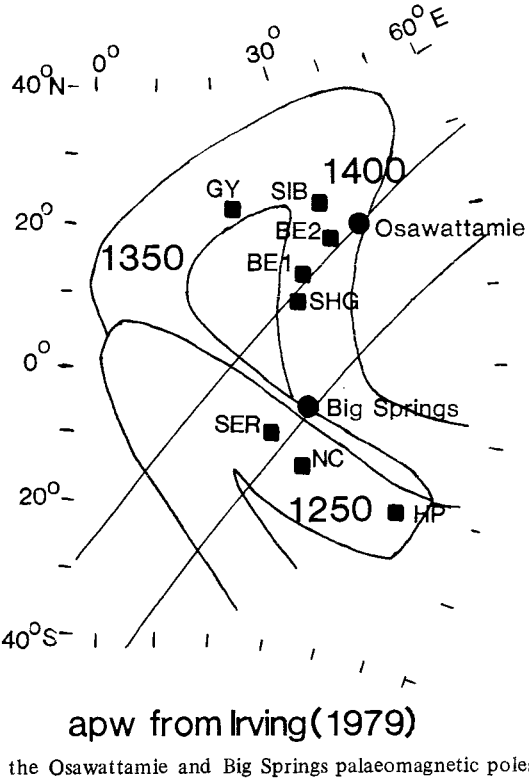


Figure 10. Location of the Osawattamie and Big Springs palaeomagnetic poles on Irving's (1979) mid-Proterozoic apparent polar wander path for Laurentia. Small circles trending NE–SW are the locus of points on which the Osawattamie and Big Springs poles could lie given just the inclination information in the cores. Other poles are from Irving (1979). SIB = Sibley Group redbeds; BE1 and BE2 = Belt Series, Alta.-Montana; SHG = Sherman granite, Colorado; SER = Seal Lake Group redbeds, NC = Nain Anorthosite Complex, Labrador; HP = Harp dykes, Labrador; GY = Grand Canyon Supergroup, Arizona.

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aligned parallel to the present field declination, the characteristic directions clustered near to $D \simeq 245°$.

Samples A1, A2 and A3 were not demagnetized until 1.5–2 yr after the core was drilled. They had been stored in random orientations for 1 yr during this period. These three samples had been oriented relative to each other when cut from the core; however, the directions of their low-temperature, low-coercivity components were different and would not align these samples' characteristic directions with the characteristic directions of samples 1, 2 and 4.

Samples A1, A2 and A3, then, could not be oriented by their low-temperature, low-coercivity magnetizations. Since sample A1 was cut from the core directly adjacent to sample 2, these two samples could be oriented relative to each other by a driller's scratch mark. Samples 2, A1, A2, A3 and 3 now had relative azimuthal orientation, but their absolute orientation was based only on the low temperature–low coercivity magnetizations of sample 2. So as not to weight too heavily the present field directions of sample 2 in calculating the final mean direction for the Big Springs core, the mean direction of samples 2, A1, A2, A3 and 3 was combined with the characteristic directions of samples 1 and 4. This hierarchical averaging scheme gave a Big Springs magnetization direction of $I = 49.0°$, $D = 249.6°$ ($K = 51.4$, $\alpha_{95} = 11.3°$, Fig. 9), agreeing with Briden & Ward's (1966) inclination estimate of 49° and having a K which lies within the error bounds of the K estimated by the Briden & Ward method ($K = 79 \pm 66$). The palaeomagnetic pole corresponding to the mean direction is located at lat. = 4.5°S, long. = 29.9°E ($\delta m = 14.9°$, $\delta p = 14.1°$, Fig. 10).

6 Discussion

The relatively high Koenigsberger ratio for the Osawattamie core ($Q_n = 2.05$) may suggest that it has a greater palaeomagnetic stability than the Big Springs core ($Q_n = 0.38$); however, this interpretation is probably oversimplified. Since this core has the largest crystalline grain size of the two cores its Q_n would be expected to be lower than the Big Springs Q_n because the Osawattamie probably contains more coarse-grained, multidomain magnetite. The average susceptibilities of the two cores are nearly identical (Osawattamie, $K = 0.00344$ SI units, Big Springs, $K = 0.00340$ SI units). The difference in Koenigsberger ratios comes entirely from the higher average NRM intensity for the Oswattamie core. This makes the Oswatammie NRM intensity suspect.

In addition to its high intensity, the Osawattamie's NRM has very steep inclinations which are removed by af demagnetization fields of 10 mT but are not removed by thermal demagnetization as high as 580°C. The Big Springs material has NRM inclinations which are close to the present field direction inclination (~68°). They are removed easily by both af and thermal demagnetization. This disparity in behaviour would suggest that the Osawattamie core acquired a remanence during drilling parallel to the core axis. The dissimilarity between the ARM and NRM af demagnetization curves (Fig. 8) further supports this interpretation, if ARM can be used as a model for thermoremanent magnetization (TRM) (Levi & Merrill 1976). Lowrie & Kent (1976) have suggested that a secondary component of magnetization seen parallel to the axis of DSDP cores from site 319 could be a VRM acquired during drilling. The VRM acquired by the Osawattamie sample shows the same characteristic behaviour that Lowrie & Kent (1976) observed for samples from DSDP sites 319 and 321. The sample was af demagnetized before VRM acquisition and showed a change in VRM acquisition rate during the experiment. However, 86 per cent of the VRM acquired by the Osawattamie sample was removed by an af field of 1.25 mT and the VRM was generally much less resistant to af demagnetization than the NRM. These results suggest

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that the vertical secondary moment in the Osawattamie core is not a VRM. The observation that the Osawattamie's secondary vertical moment was easily removed by af demagnetization but not by thermal demagnetization would suggest that it is an IRM. It is not known what the magnitude of the field in the drill core apparatus was, but it was probably no greater than 10 mT; however, the 10 mT IRM was much less resistant to af demagnetization than the NRM (Fig. 8). It was not until the sample acquired a near saturation IRM of 50 mT that the demagnetization curves of NRM and IRM start to become similar; however, it would seem unlikely that the Osawattamie samples acquired a saturation IRM in a field $\geqslant 50$ mT, parallel to the core axis during drilling.

One possible explanation for the secondary vertical component in the Osawattamie core is that it is a piezoremanent magnetization (PRM). Carmichael (1968) has shown for single crystal magnetite rods that the resistance of PRM to af demagnetization is greater than that of an IRM. In addition, the PRM is generally larger than the IRM for the same direct field. If the Osawattamie core acquired a PRM during drilling it would explain why the overprint behaves as an IRM but is much more resistant to af demagnetization than IRMs which are experimentally induced in fields that might be expected to be caused by the drilling apparatus. The presence of the PRM does not invalidate the use of the Osawattamie low coercivity–low temperature component to orient the core azimuthally since a PRM applied exactly parallel to the core's axis would not affect the horizontal component of the low coercivity–low temperature present field magnetization. Declination information would still be retrieveable; however, its accuracy may be decreased if the PRM was not applied exactly parallel to the core axis.

If the Osawattamie high NRM intensity is due primarily to a PRM overprint, its true Q_n is probably closer to that of the Big Springs core. This low Q_n (~ 0.38) is more consistent with the medium to coarse crystalline grain size for these rocks which suggests a large multidomain component in the NRM. However, the resistance of a small part of the remanence (1–5 per cent) to af demagnetization fields as high as 60–80 mT for both cores, would suggest that the characteristic direction is carried by a very fine-grained magnetic fraction which has high palaeomagnetic stability.

The declination of the low temperature–low coercivity component was aligned parallel to the present field declination ($D = 8°$) rather than the axial dipole declination ($D = 0°$) to orient these cores for the following reason: Specimens which were demagnetized 4 months after the core was drilled had recognizable present field, low-temperature, low-coercivity magnetization components; specimens demagnetized nearly 2 yr later did not. This would suggest that the low-temperature, low-coercivity components have relaxation times of only 1 yr and they are more properly identified with present field overprints. The very low coercivity component (< 2.5 mT) seen in some samples (A-1-1 and A-3-1) is probably a viscous magnetization picked up during storage.

No tilt corrections were made to the characteristic directions when palaeomagnetic poles were calculated since the early Palaeozoic section directly above the Precambrian granitic basement has dips $< 1°$, indicating no major tectonic events in the Phanerozoic. Major tilting probably did not occur in the Precambrian since the granitic intrusives are undeformed and have been interpreted as having been emplaced in an anorogenic environment (Van Schmus & Bickford 1981).

The palaeomagnetic poles calculated for the Big Springs core and the Osawattamie core are nearly 25° apart yet their radiogenic ages differ only by 22 Ma. If the pole positions and U-Pb ages are correct, this would necessitate a very high rate of polar wander at this time. It is more likely that the orientation technique was not totally accurate and that the poles should fall closer together. The small circles calculated from the inclination data are only

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12° apart which could reduce the required polar wander rate. It is likely that the Osawattamie core's declination value may be slightly in error due to the PRM overprint the core probably acquired during drilling. Since the Big Springs core apparently did not acquire a PRM its pole position is probably closer to the value that would be obtained from a conventionally oriented core.

The Big Springs palaeomagnetic pole (1339 Ma) falls near several poles of the same age, the Seal Lake Group red beds (~1350 Ma) (Roy & Fahrig 1973), the Nain Anorthosite Complex, Labrador (~1400 Ma) (Murthy 1978), and the Harp dykes, Labrador (~1350 Ma) (Irving, Emslie & Park 1977) (Fig. 10).

The Osawattamie pole (1361 Ma) should probably lie closer to the Big Springs pole for the reasons discussed above; however, it does fall close to several poles of nearly the same age, the Belt Series poles (BE1 and BE2 of Irving 1979) which are estimated to be 1400 Ma, the Sibley Group red beds (1370 Ma) (Robertson 1973), the Grand Canyon Supergroup (~1380 Ma) (GY of Irving 1979) and the Sherman granite, Colorado (~1410 Ma) (Eggler & Larson 1968).

If Irving's (1979) polarity interpretation for the Proterozoic apw path is correct, then the Osawattamie core carries a normal polarity direction and the Big Springs core is reversed polarity. The presence of polarity reversals gives added support to the assumption that the *characteristic directions* are primary.

7 Conclusions

This study demonstrates that it can be possible to obtain reasonably accurate palaeomagnetic poles from azimuthally unoriented core using a technique first described by Van der Voo & Watts (1978). Apparently if demagnetization studies are conducted soon after drilling the low-coercivity–low-temperature components of magnetization will allow a maximum amount of palaeomagnetic information to be extracted from partially oriented cores. The data could aid in the age dating and correlation of the core material and would be an important supplement to data typically collected on unoriented cores.

Acknowledgments

Don Steeples of the Kansas Geological Survey (KGS) originally conceived of this study. Discussions with Harold Yarger, also of KGS, helped to clarify my thinking on several points during analysis of the data. This project was supported, in part, by funds from the Kansas Geological Survey.

References

Briden, J. C. & Ward, M. A., 1966. Analysis of magnetic inclination in bore cores, *Pure appl. Geophys.*, **63**, 133–152.

Carmichael, R. S., 1968. Remanent and transitory effects of elastic deformation of magnetite rods, *Phil. Mag.*, **17**, 8th Ser., 911–927.

Denison, R. E., Lidiak, E. G., Bickford, M. E. & Kisvarsany, E. B., 1982. Geology and geochronology of Precambrian rocks in the central interior region of the United Stated, *Econ. Geol.*, in press.

Eggler, D. M. & Larson, E. E., 1968. Palaeomagnetic study of late Precambrian rocks of the Front Range, Colorado–Wyoming, *Geophys. J. R. astr. Soc.*, **14**, 497–504.

Irving, E., 1979. Paleopoles and palaeomagnetism of North America and speculations about displaced terrains, *Can. J. Earth Sci.*, **16**, 669–694.

Irving, E., Emslie, R. F. & Park, J. K., 1977. Palaeomagnetism of the Harp Lake Complex and associated rocks, *Can. J. Earth Sci.*, **14**, 1187–1201.

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Kean, W. F. & Swingen, R. A., 1976. Palaeomagnetism of a Keweenawan age intrusive near Clam Lake, Wisconsin, *J. geophys. Res.*, **81**, 4173–4184.

Levi, S. & Merrill, R. T., 1976. A comparison of ARM and TRM in magnetite, *Earth planet. Sci. Lett.*, **32**, 171–184.

Lowrie, W. & Kent, D. V., 1976. Viscous remanent magnetization in basalt samples, in *Init. Rep. Deep Sea drill. Proj.*, **34**, 479–484, eds Yeats, R. S., Hart, S. R. *et al.*, Washington, DC, US Government Printing Office.

Murthy, G. S., 1978. Palaeomagnetic results from the Nain anorthosite and their tectonic implications, *Can. J. Earth Sci.*, **15**, 516–525.

Ocola, L. C. & Meyer, R. P., 1973. Central North American Rift, System 1 Structure of the axial zone from seismic and gravimetric data, *J. geophys. Res.*, **78**, 5173–5194.

Peirce, J. W., 1976. Assessing the reliability of DSDP palaeolatitudes, *J. geophys. Res.*, **81**, 4173–4184.

Robertson, W. A., 1973. Pole position from thermally cleaned Sibley Group sediments and its relevance to Proterozoic magnetic stratigraphy, *Can. J. Earth Sci.*, **10**, 180–193.

Roy, J. L. & Fahrig, W. F., 1973. Palaeomagnetism of Seal and Croteau rocks from the Grenville Front, Labrador: polar wandering and tectonic implications, *Can. J. Earth Sci.*, **10**, 1279–1301.

Steeples, D. W., 1980. Microearthquakes recorded by the Kansas Geological Survey, *J. Kans. geol. Surv.*, **2**, 14.

Steeples, D. W. & Bickford, M. E., 1981. Piggyback drilling in Kansas: An example for the continental drilling program, *Eos*, **62**, 473–476.

Van der Voo, R. & Watts, D. R., 1978. Palaeomagnetic results from igneous and sedimentary rocks from the Michigan Basin borehole, *J. geophys. Res.*, **83**, 5844–5848.

Van Schmus, W. R. & Bickford, M. E., 1981. Proterozoic chronology and evolution of the mid-continent region, North America, in *Precambrian Plate Tectonics*, pp. 261–296, ed. Kroner, A., Elsevier, New York.

Yarger, H. L., 1981. Aeromagnetic survey of Kansas, *Eos*, **62**, 173–178.

Zijderveld, J. G. A., 1967. AC demagnetization of rocks: analysis of results, in *Methods of Palaeomagnetism*, pp. 254–286, eds Collinson, D. W., Creer, K. M. & Runcorn, S. K., Elsevier, New York.

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